Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/16/2015
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Project Title:	Signaling at the Vestibular Calyx Synapse		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80045	**Congressional District:**	6
Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: End date changed to 3/31/2015 per NSBRI (Ed., 11/11/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rennie, Katherine (MENTOR/ University of Colorado, Denver)		
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Task Description:	POSTDOCTORAL FELLOWSHIP The vestibular system is responsible for our sense of balance. Precisely how mechanical signals generated by head movements are detected and transformed into electrical signals by hair cells and their afferent nerve fibers is poorly understood. Signals are altered during spaceflight, changing the perception of body orientation and movement. Astronauts experience dizziness and disorientation during the first few days in microgravity and again on return to Earth. Temporary disorientation on leaving and on returning suggests that there is plasticity in the system. Our goal is to better understand the transfer of information between hair cells and afferents as well as the mechanisms underlying plasticity. This is important in understanding alteration of vestibular perception in space. Mammalian vestibular epithelia can be divided into central and peripheral zones which differ in hair cell morphology, ribbon synapse number, afferent innervation, and firing pattern. Calyx-only afferents contact type I hair cells in central regions, whereas dimorphic afferents form bouton and calyx endings and supply peripheral and central hair cells. Action potential firing is irregular in calyx-only afferents and varies from regular to irregular in dimorphic afferents. We are interested in how mechanisms on either side of the type I hair cell/calyx synapse shape transmission in central and peripheral zones. Our first aim was to use whole cell patch clamp to record postsynaptic responses from semicircular canal calyx afferents. Our second aim was to investigate efferent modulation of calyx responses. Key findings: 1. To investigate the hypothesis that zonal variations in calyceal ion channels play a role in shaping postsynaptic events, I developed a slice preparation of the gerbil crista to record spontaneous excitatory postsynaptic currents, action potentials, and voltage-dependent conductances from central zone (CZ) and peripheral zone (PZ) calyx terminals. Spontaneous firing was regular or irregular and frequency ranged from 1.3 to 63 Hz. Firing persisted when transduction channels in hair cell stereocilia were blocked by streptomycin and when postsynaptic receptors were blocked by NBQX, suggesting that calyceal ion channels play a role in generating action potentials. 2. We hypothesized that zonal variations in ion channel expression underlie the variation in inactivation of potassium currents previously seen in isolated calyces. I found that inactivation, expressed as percent reduction of peak current, had a median value of 9.4% in PZ calyces (n=66), which was significantly different from the value of 4.2% in CZ calyces (n=33; $P < 0.001$, Mann Whitney Rank Sum Test). The greater inactivation of potassium currents in PZ calyces could contribute to tonic activity of regular afferents, which may be better suited to signaling long duration stimuli. 3. We investigated the contribution of diverse potassium channels to whole cell outward current. XE991 dihydrochloride (20 µM), a KCNQ channel antagonist, reduced outward currents in both CZ and PZ calyces. Mean reduction in PZ calyces was 31.4% (±3.59%, SEM; n=10) which was significantly greater than the value of 17.6% (±3.3%, SEM, n=8; $P = 0.014$) seen in CZ calyces. XE991 also reduced the frequency of spontaneous action potentials. 4-aminopyridine (10-50 uM) blocked rapidly activating and moderately inactivating currents more prevalent in PZ calyces. Dendrotoxins, which block Kv1.1, 1.2, and 1.6 channels, reduced a non-inactivating current in CZ calyces and dampened excitability. Dendrotoxin-sensitive currents may allow detection of high-frequency signals and may therefore be important in generating rapid vestibular-driven reflexes. Margatoxin, a Kv1.3 and 1.6 channel antagonist, blocked a slowly inactivating current more prevalent in PZ calyces. These findings have been detailed in two recently published papers, one in the Journal of Neurophysiology and one in Frontiers in Systems Neuroscience.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Currently there are many therapeutic interventions helpful in treating dizziness, vertigo, and space motion sickness. However, their mechanism of action is mostly unknown. This work investigated cellular mechanisms underlying balance sensation in rodents. A better understanding of how sensory information is transformed by vestibular hair cells into neural signals will elucidate signaling mechanisms in a normal environment and under conditions such as spaceflight, where gravity is altered. This work has contributed to our understanding of how variations in KCNQ and Kv1 ion channels in calyces supplying peripheral and central zones of the vestibular sensory epithelium shape signaling. This could drive future studies incorporating pharmacological interventions to combat vestibular dysfunction.
Task Progress:	During the first year of funding, I developed a novel crista slice preparation so that we could investigate regional differences in signaling in calyx terminals supplying central and peripheral zones of the sensory epithelium. I found a significant difference in the inactivation properties of the macroscopic whole cell outward current. Inactivation was greater in calyces innervating peripheral regions of the crista compared to those innervating central zones. During the second year of funding, I explored underlying differences in potassium channels that could explain the difference in inactivation properties. I did this by testing the effect of various K+ channel blockers on the whole cell outward current and on firing properties. I found higher expression of moderately inactivating K+ currents in peripheral calyx terminals that were sensitive to 4-aminopyridine (a broad K+ channel antagonist) and to margatoxin which targets Kv1.3 and 1.6 channels. Non-inactivating currents sensitive to dendrotoxin, which blocks Kv1.1 and 1.2 K+ channels, were prevalent in central zones of the crista. I also applied the slice technique to the utricle and found that, similarly to the crista, K+ current inactivation was greater in the peripheral (extrastriolar) zone compared to the striolar (central) zone. The presence of more inactivating currents peripherally could contribute to tonic firing characteristics, making these afferents better suited to signal long duration stimuli. Dendrotoxin-sensitive, low-voltage-activated currents centrally could underlie more phasic, irregular responses, suitable for signaling rapid changes in head position. These findings were recently published in 2 papers in peer-reviewed journals.
Bibliography Type:	Description: (Last Updated: 10/14/2015)
Articles in Peer-reviewed Journals	Meredith FL, Kirk ME, Rennie KJ. "Kv1 channels and neural processing in vestibular calyx afferents." Front Syst Neurosci. 2015 Jun 2;9:85. http://dx.doi.org/10.3389/fnsys.2015.00085 , Jun-2015
Articles in Peer-reviewed Journals	Meredith FL, Rennie KJ. "Zonal variations in K+ currents in vestibular crista calyx terminals." Journal of Neurophysiology. 2015 Jan 1;113(1):264-76. Epub 2014 Oct 15. http://dx.doi.org/10.1152/jn.00399.2014 ; PubMed PMID: 25343781 , Jan-2015